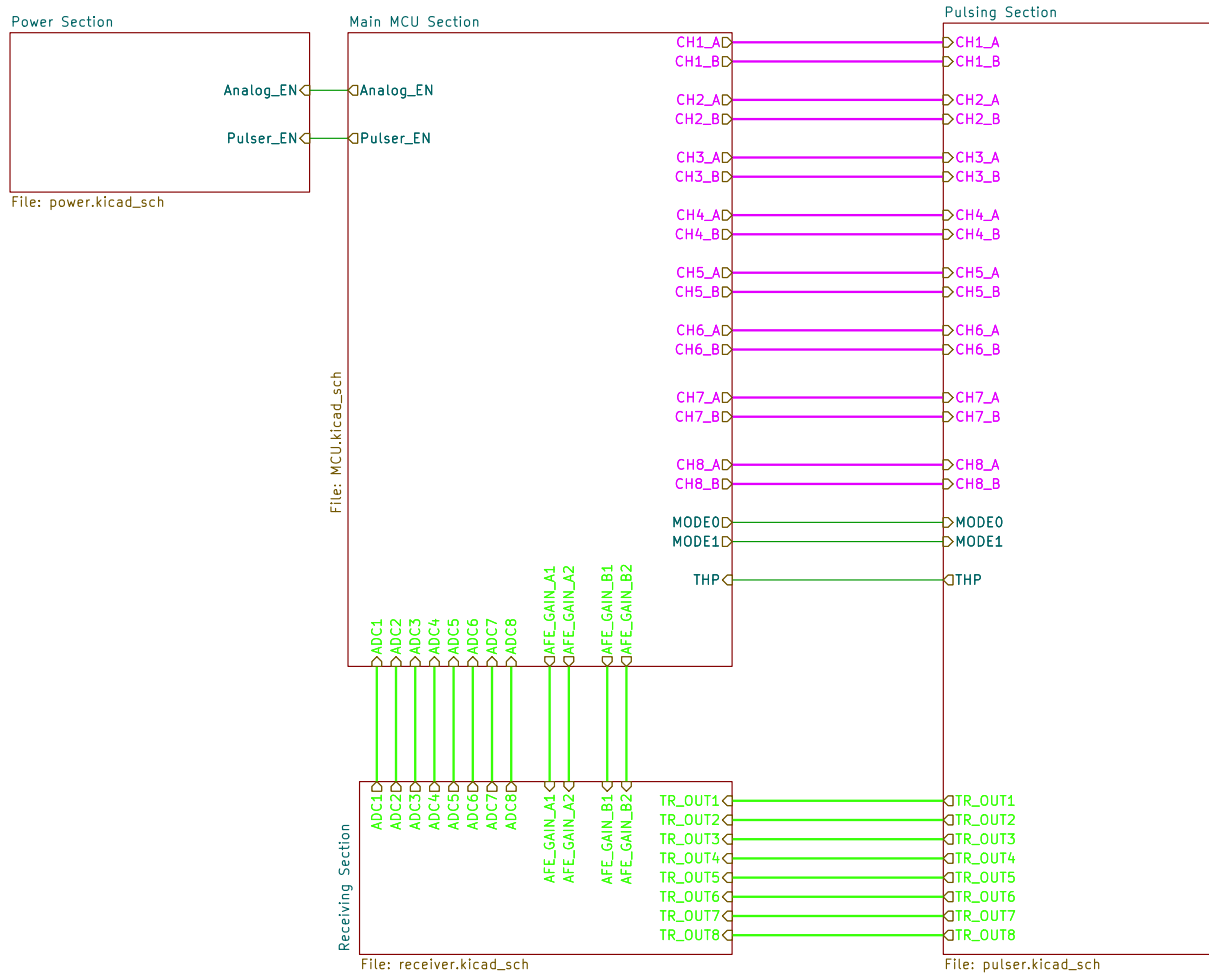
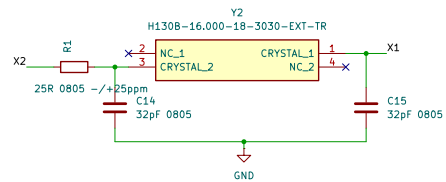
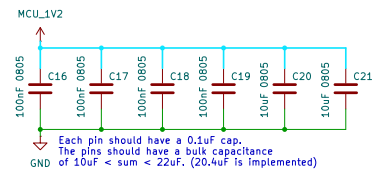
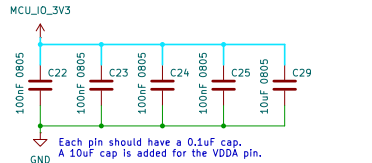




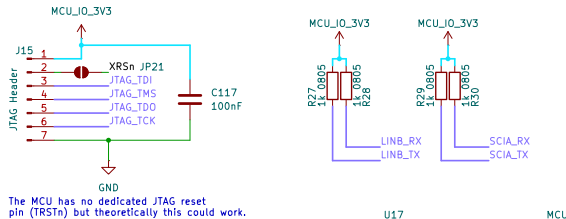
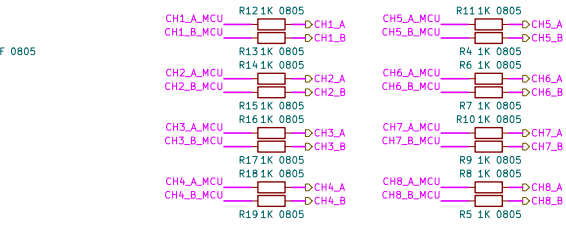
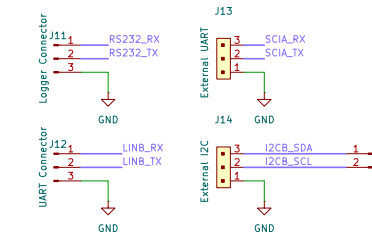
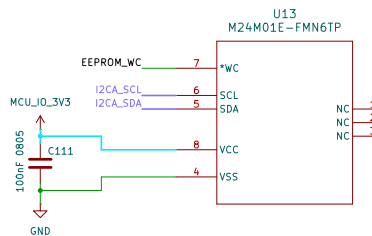
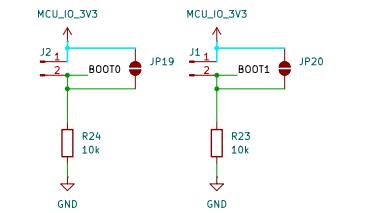
12 Volt (External) (555mA)	7 Volt (Buck Converter) (350mA)	5 Volt (LDO Regulator) (200mA)	LNA + VGA Supply (160mA – short circuit)
			Op-Amps (4 * 10mA = 40mA)
	5 Volt (Buck Converter) (205mA)	3.3 Volt (LDO Regulator) (150mA)	MCU Supply (150mA)
		Positive Pulse Driver Supply (100mA)	
		3.3 Volt (LDO Regulator) (5mA)	Pulser Logic Supply (5mA)
		-5 Volt (Inverting Buck Converter) (100mA)	Negative Pulse Driver Supply (100mA)



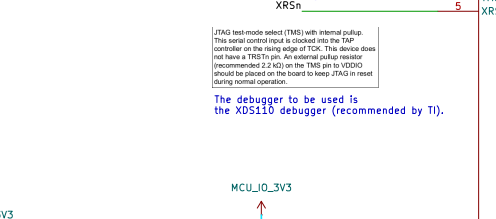
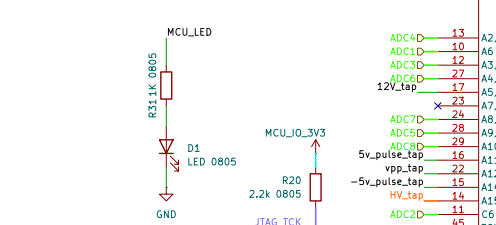
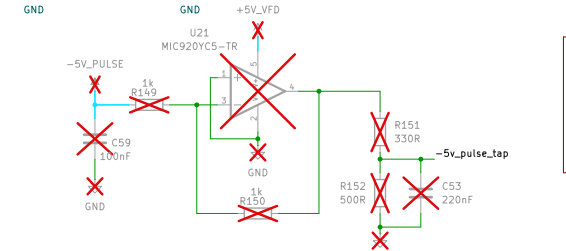
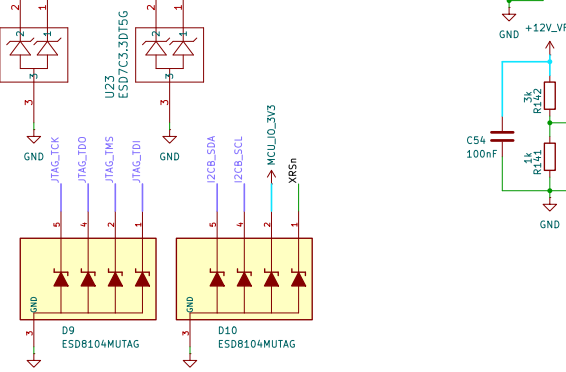
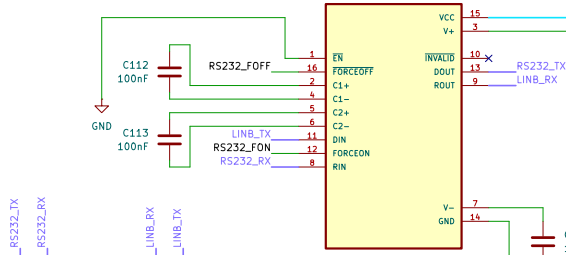
$C1 = C2 = 2(C_{load} - C_{stray}) = 2(18 - 2) = 32pF$
 The crystal is required ($\approx 16MHz$):
 - To have a drive level of $\geq 1mW$ (a damping resistor is used)
 - Have maximum ESR of 75 Ohms (50 is used)
 - $12pF < C1 < 24pF$ (18pF is used)



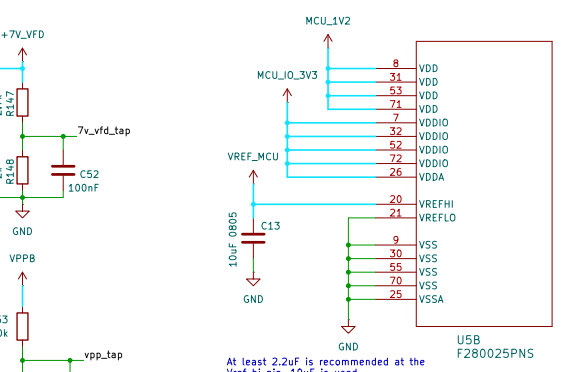
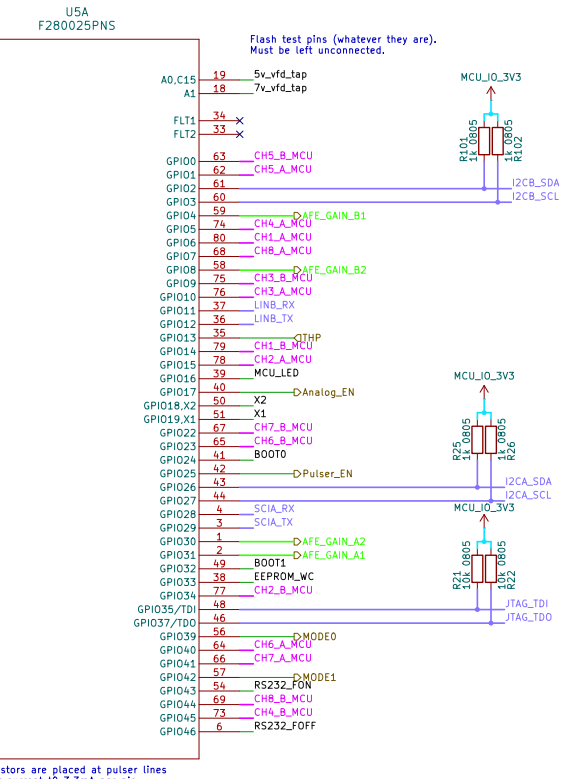
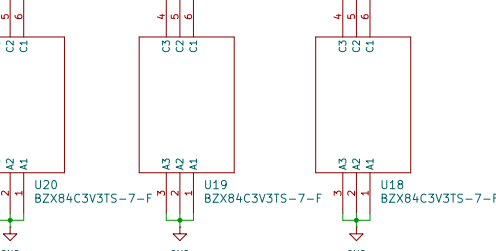
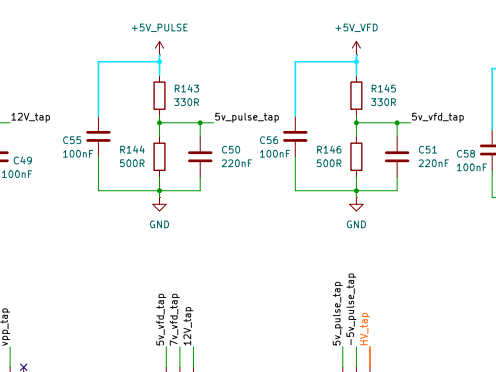
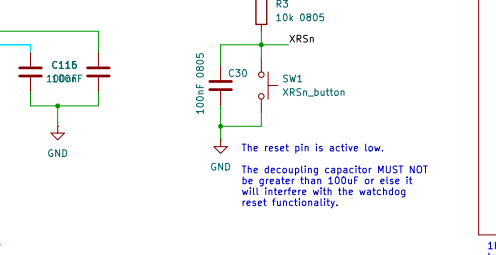
BOOT MODE	SPICOR (DEFAULT) BOOT MODE SELECT	SPICOR (DEFAULT) BOOT MODE SELECT
Pin#10	0	0
SPICOR	0	1
CAN	1	0
Flash	1	1

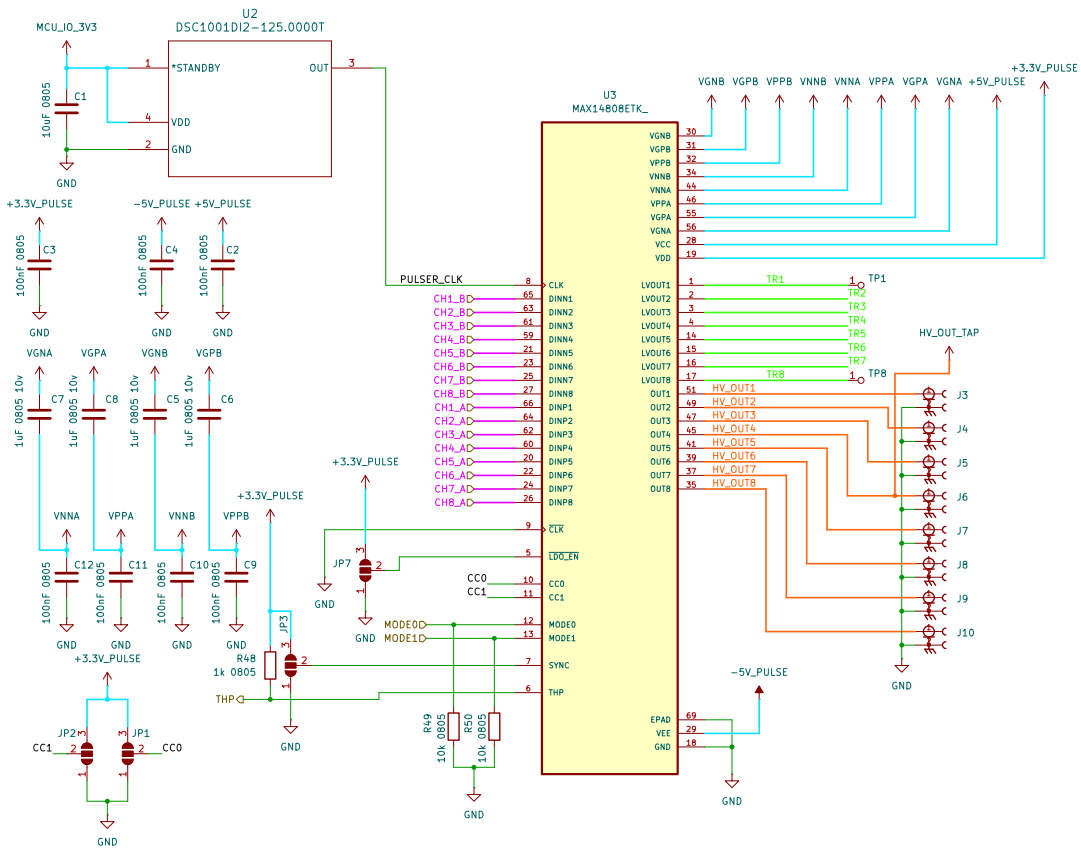


The MCU has no dedicated JTAG reset pin (TRSTn) but theoretically this could work.



The MCU has no dedicated JTAG reset pin (TRSTn) but theoretically this could work.





VGXX: gate driver voltage supplies
(positive and negative for ports A & B)

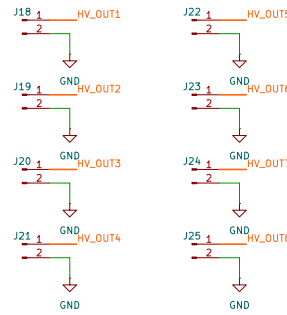
VNNX/VPPX: positive and negative HV supplies
(for ports A & B)

Enabling/Disabling the Internal floating power supplies (FPS):
requirements in most of the cases. Drive LDO_EN low
or leave unconnected to enable the internal FPSs; drive
LDO_EN high to disable the internal FPSs.

A triple solder jumper is placed
to either pull-up (select external FPS)
or pull-down (select internal FPS) the
LDOEN pin.

Table 5. Current Drive Selection

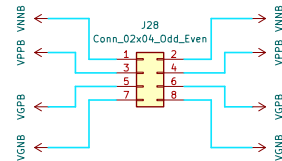
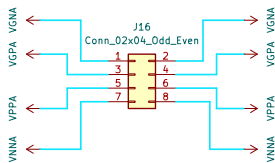
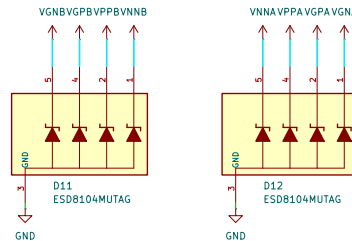
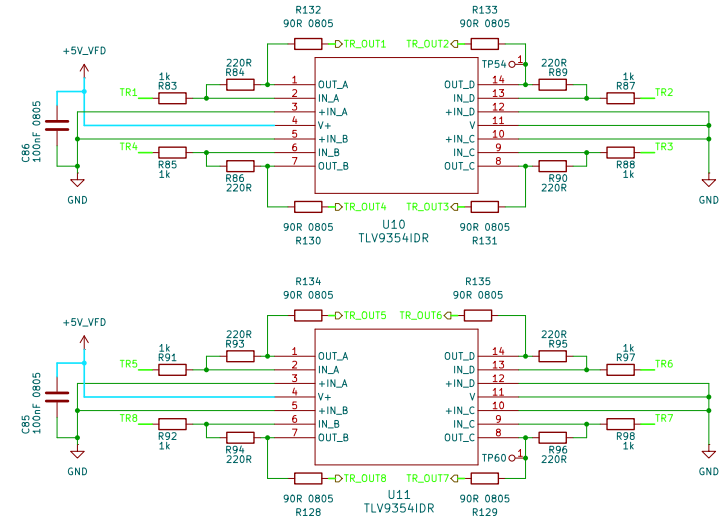
INPUTS		PULSER OUTPUT CURRENT (typ)
CC0	CC1	
0	0	2A
1	0	1.5A
0	1	1A
1	1	0.5A

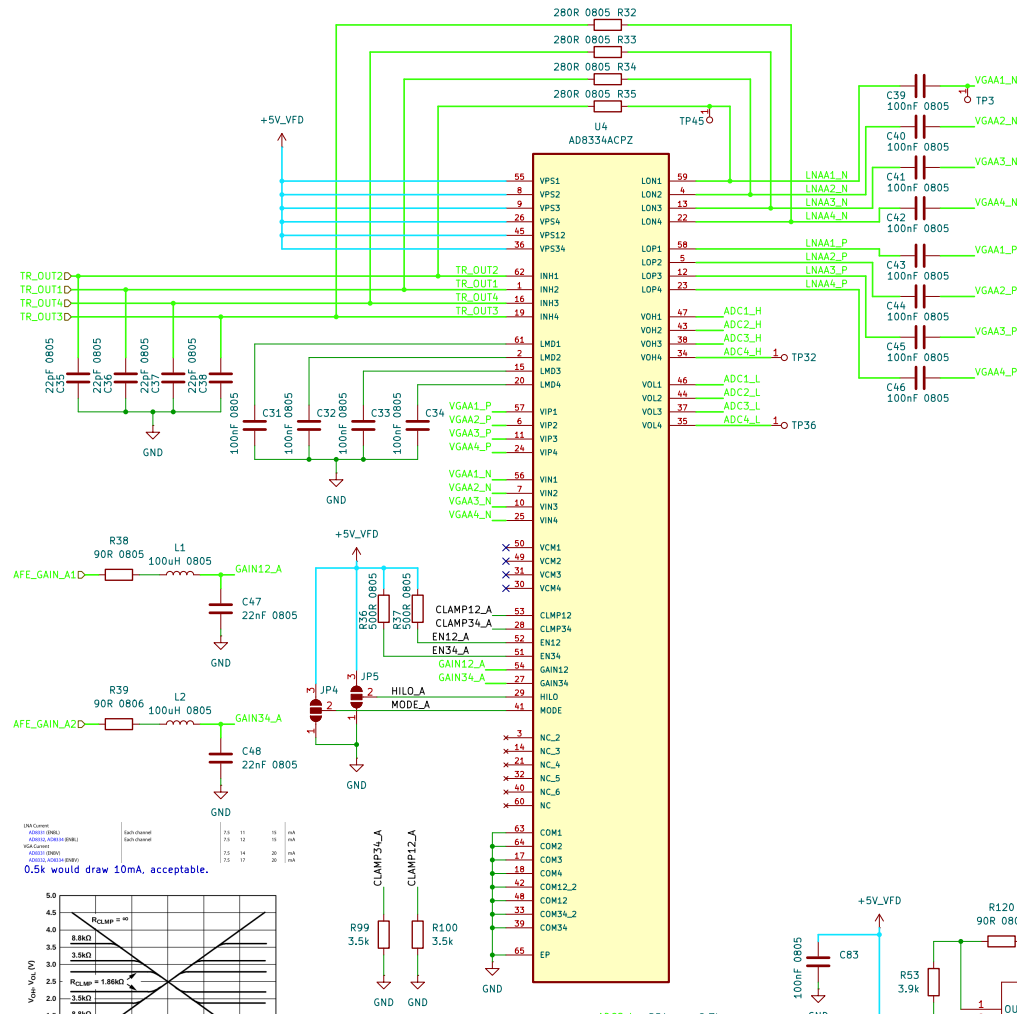


Output swing from the pulser:
-1.2 -> 1.2 (2.4v swing)

Input voltage range for the LNA+VGA:
-0.275 -> 0.275 (0.55v swing)

The differential opamp configuration
produces a gain of 0.22 for a swing
of 0.528v





Load Current	Each Channel	Each Channel
AD8334 (RMS)	7.5	11
AD8334 (RMS) (RMS)	7.5	11
AD8334 (RMS)	7.5	11
AD8334 (RMS) (RMS)	7.5	11
AD8334 (RMS)	7.5	11
AD8334 (RMS) (RMS)	7.5	11

0.5k would draw 10mA, acceptable.

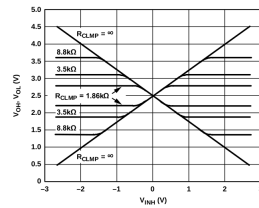
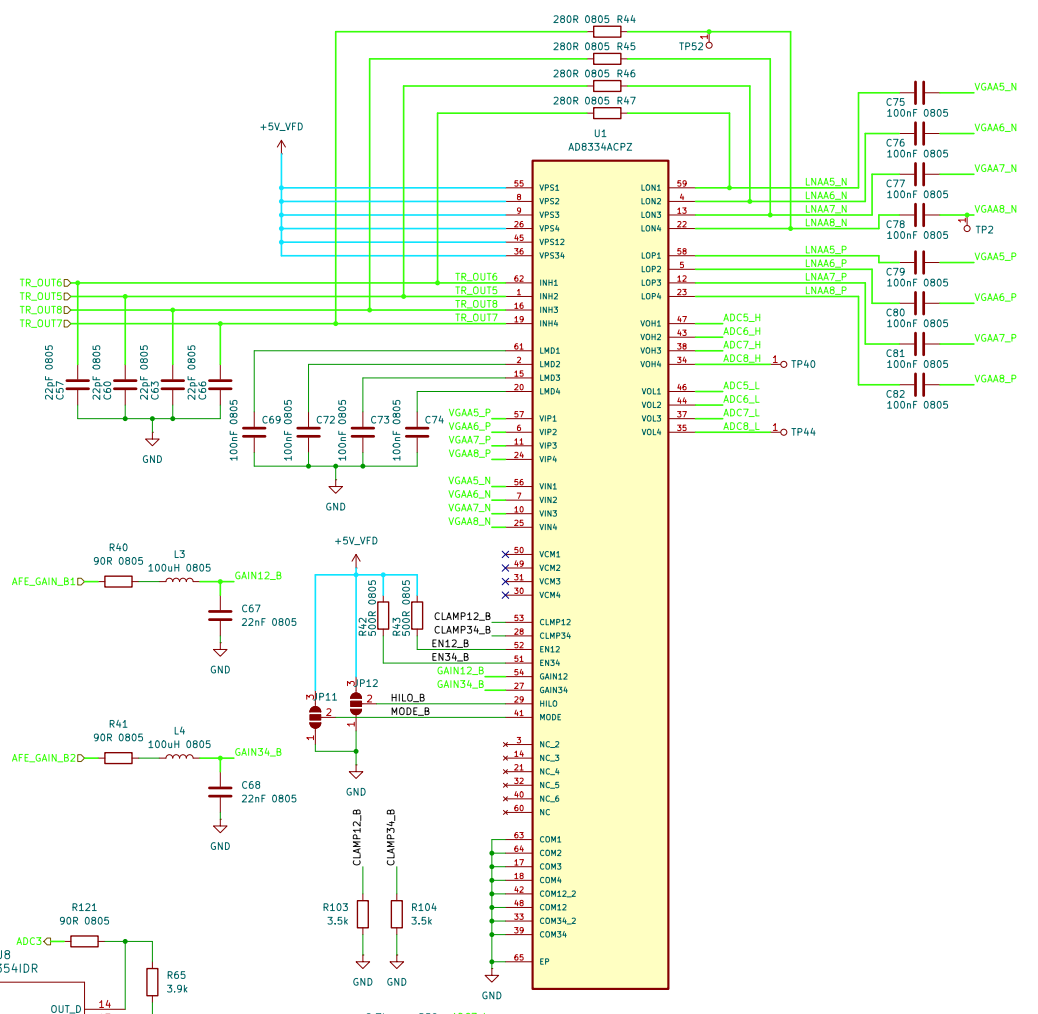


Figure 80. Output Clamping Characteristics

Table 7. LNA External Component Values for Common Source Impedances		
R _{in} (Ω)	R _{in} (Closest STD 1% Value, Ω)	C _{in} (pF)
50	280	22
75	412	12
100	560	8
200	1.13 k	1.2
500	3.01 k	None
6k	∞	None

2nd order filter values were recommended by a TI application note (spraa88a), specs:
 - PWM frequency = 5 MHz
 - Cut-off frequency = 107.3 KHz
 - Damping ratio = 0.675 (over-damped but response time = 3 uSec)

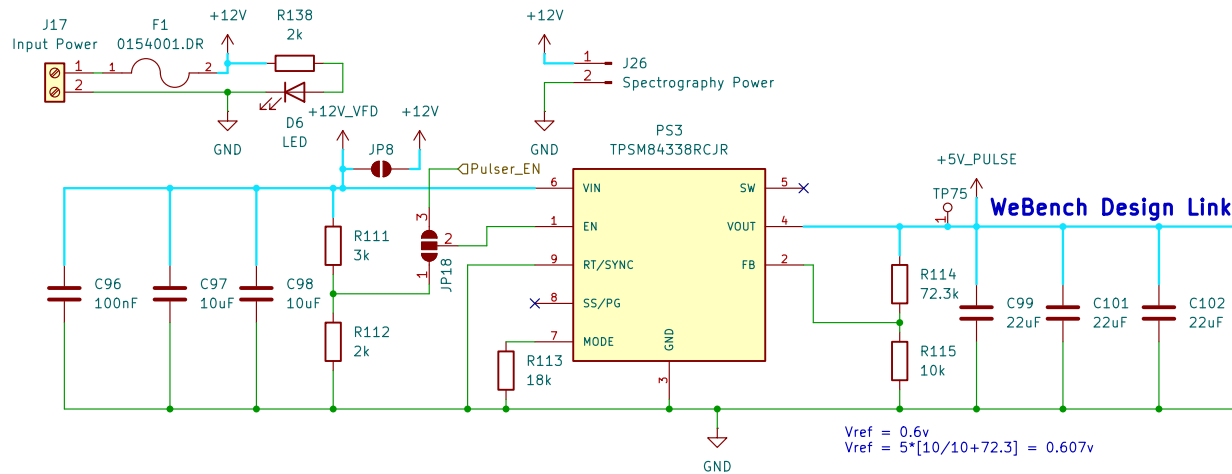
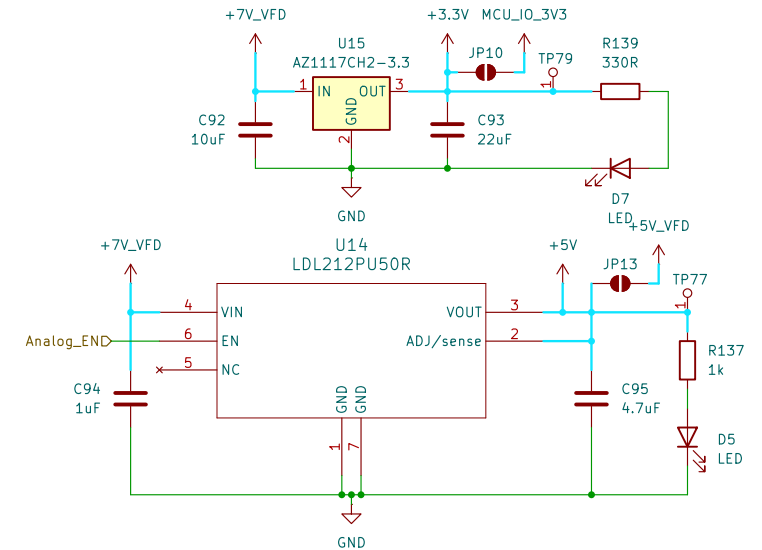
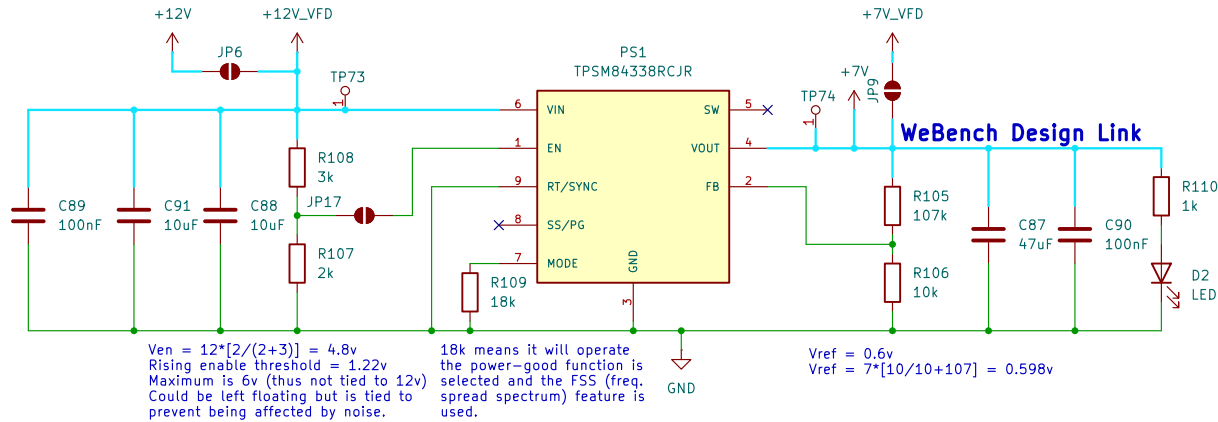
MODE INTERFACE (PIN MODE)		
Logic Level for Positive Gain Slope	0	1.0 V
Logic Level for Negative Gain Slope	2.25	5 V
HELD GAIN RANGE INTERFACE (PIN HELD)		
Logic Level to Select HI Gain Range	2.25	5 V
Logic Level to Select LO Gain Range	0	1.0 V



Output voltage from LNA+VGA:
 -1.125 → 1.125 (2.25v swing)
 Input voltage range for the MCU's ADC:
 0 → 3.3 (3.3v swing)
 The differential opamp configuration shifts the signal and amplifies it with a factor of 1.444 (3.9/2.7) to achieve a swing of 3.25.

VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.

In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the *VDD Decoupling* section for VDD decoupling configurations.



The LC-filter at Vin is recommended by the datasheet.

